

S1 Appendix

Early longitudinal trajectories and rank persistence of published blood RNA signatures in sepsis: a multicohort transcriptomic study

Xinzhao Li, Zihao Yan, Xinyue Tu and Yi Gong

This file contains expanded methods, Tables S1–S13 and legends for Figures S1–S6. Machine-readable results, cohort registries and supporting analyses are supplied in S1 Data; exact source tables for the main and supplementary figures are supplied in S2 Data; reproducible analysis code is supplied in S1 File.

Supplementary Methods

Public dataset identification and search update

Public longitudinal transcriptomic datasets were evaluated using prespecified eligibility rules: a source-defined human sepsis or septic-shock population, blood-derived transcriptomic measurements, a recoverable patient-level baseline, repeated patient-level sampling within a prespecified window, and sufficient gene coverage for fixed-signature reconstruction. Searches of GEO, BioStudies/ArrayExpress, ENA, dbGaP, Zenodo and PubMed were refreshed through 20 July 2026 using concepts for sepsis, septic shock, blood, transcriptomics, serial or longitudinal sampling, repeated or paired measurements and time points.

Four candidates identified by the update underwent formal adjudication. GSE106878 supplied 47 public baseline-to-24-h pairs and complete coverage of all 74 genes and was retained as a separate directional-replication cohort. E-MEXP-3850 lacked public expression files; PRJEB111201 had public raw reads but no processed subject-by-gene matrix compatible with the established workflow; and phs003608.v1.p1 lacked an openly accessible subject-by-gene bulk-expression matrix. None was added to the primary six-cohort synthesis. The broader machine-readable registry is retained in S1 Data.

Signature identification and reproducibility eligibility

A structured pre-performance source review registered 34 candidate signatures and froze intended-use strata and A1/A2/B/X reproducibility grades on 15 July 2026, before longitudinal outcomes were inspected. Eligibility required a publication-time fixed gene set and scoring rule, a defined direction, and application without feature selection or model refitting. Formula evidence was checked against primary articles, formal supplements, author code, patents or official technical sources.

Ten candidates were graded A2, 19 B and five X; none was A1. The panel comprised all eight protein-coding A2 signatures with resolved cross-platform mapping. The remaining A2 candidates, SIG010 and SIG032, were long noncoding RNA panels with unresolved cross-platform mapping. No selection decision used longitudinal trajectory, effect magnitude, clinical association or statistical significance. This was a structured reproducibility audit rather than an exhaustive systematic review.

Cohort independence and signature-specific eligibility

The study family was the unit of independence. GSE13904 shared patient tokens with GSE8121 and was not counted as a separate family. GSE131411 was treated as a reanalysis or extension of GSE110487. GSE54514 had possible same-program overlap with the development background of SeptiCytE LAB, *FAIM3:PLAC8* and sNIP. It remained available for descriptive clinical context but was removed from the primary-independent pooled estimates for these formulas. Cross-signature concordance required eligibility for both members of a pair.

GSE106878 provided a separate T24 cohort for directional replication and did not enter the six-cohort temporal meta-analysis. Its randomized hydrocortisone and placebo arms were retained [1, 2]. Any 28-day mortality analysis included treatment assignment as a covariate.

Clinical metadata mapping

Every analyzed clinical sample required a resolved sample-to-patient, time-point and clinical-status link. GSE95233 had 51 complete admission-to-second-measurement pairs, including 20 D2 and 31 D3 samples. The deposited 28-day status identified 34 survivors and 17 nonsurvivors. Combined models adjusted for D3 versus D2, and day-stratified analyses were prespecified sensitivities.

GSE110487 had 31 complete pairs. The source definition classified a patient as a nonresponder when $\text{SOFA}(T1) - \text{SOFA}(T2) < 5$ and $\text{SOFA}(T2) > 8$; all others were responders. A deposited time-point field conflicted with the sample title, source name and raw-count column for 21 of 62 samples. The three mutually consistent identifiers were used. No expression value or clinical outcome was used to resolve the discrepancy.

GSE57065 provided patient-level SAPSII-High and SAPSII-Low labels for all 28 patients. The source threshold separated values above 45 from values below 45, but the individual numerical scores were not public. This variable was treated as baseline severity context rather than recovery or mortality. GSE54514 had 26 survivors and nine nonsurvivors at baseline. Available paired numbers fell to 24/7 at T24 and 21/7 at T48. Later samples were displayed with numbers at risk. One nonsurvivor day-4 sample lacked a baseline patient link and was excluded from baseline-defined paired analyses.

Rank persistence

Average ranks were assigned within each complete-pair cohort-signature-landmark set. Percentile rank was $(\text{rank} - 1) / (n - 1)$. Spearman correlation summarized ordering, and the median absolute percentile change summarized movement. Each bootstrap resampled patients and recalculated both rank vectors before the two metrics were obtained. Two thousand replicates were required, with at least 95% finite results.

Fisher-z random-effects synthesis was performed only at $k \geq 3$. Restricted maximum likelihood estimated between-cohort variance. Standard Hartung-Knapp and modified Hartung-Knapp intervals were calculated [3]. When $q < 1$, the modified analysis used $q^* = 1$. The four diagnostic signatures formed one multiplicity family within each landmark.

Clinical contrast models

For a two-visit cohort, the primary outcome was patient-level ΔZ . HC3 regression estimated the adverse-minus-favorable difference. A bootstrap resampled patients within clinical status, preserving the observed group sizes. Follow-up ANCOVA adjusted for the baseline standardized score. Two-time-point random-intercept models were fitted as a consistency check. Since several random-effect variances approached zero, their P values were not used for headline inference.

GSE54514 baseline-to-T48 was its main source-defined survival contrast; T24 was secondary. Days 4 and 5 were descriptive because only five and three nonsurvivors remained. Clinical endpoints were not synthesized across cohorts. Their definitions covered 28-day survival, source-defined organ-function course, baseline severity and survival of unspecified horizon.

GSE106878 reconstruction and matched-gene benchmark

The GSE106878 series matrix and the full GPL10295 table were obtained directly from GEO. Platform symbols were reconciled to the study gene registry, and multiple unambiguous probes were averaged. Seventy-four required genes were present. The matrix contained 47 pre-treatment and 47 post-24-h samples. Every fixed formula was calculated on the deposited positive log₂ quantile-normalized values. Patient-level calculations reproduced the T24 means reported in the manuscript.

The background universe comprised 18,971 genes with unambiguous platform symbols, complete expression and positive values. Baseline mean and variance were separately divided into deciles. For each component, a replacement was drawn from the same two-dimensional bin after excluding all published signature genes and replacements already selected in the same pseudo-formula. The original coefficients, signs, geometric-mean blocks, ratios and score orientation were retained. All 8,000 iterations succeeded and every component used an exact bin match.

GSE106878 treatment-arm sensitivity

The four diagnostic signatures were summarized separately in 23 placebo and 24 hydrocortisone recipients using 2,000 patient-level bootstrap replicates with seed 20260720 and the common signature-specific baseline standard deviation from all 47 patients. Ordinary least squares modeled ΔZ as a function of a hydrocortisone indicator with placebo as reference. HC3 robust covariance provided large-sample normal confidence intervals and P values, with Holm correction across four treatment coefficients. This sensitivity evaluated dependence of direction on treatment arm and was not an efficacy or causal treatment-effect analysis.

Exact contribution identities

Linear score changes were decomposed directly. Shapley values were calculated by enumerating every coalition of component genes [4]. For the Severe-or-Mild score, exact Owen values respected the four published modules [5]. Hybrid coalition inputs used only the observed baseline and follow-up expression vectors. For every patient-signature-window record, the sum of oriented standardized contributions equaled ΔZ . All 3,344 identities met the 1×10^{-8} tolerance.

The cohort-mean leader was the gene maximizing the absolute mean signed contribution. This differs from the most common patient-level leader. Ties within 1×10^{-12} were retained as sets. Recurrence remained descriptive because only three to five independent cohorts contributed.

Missingness and biological context

Missing follow-up was defined only when the cohort scheduled the landmark and a baseline score was available. A design-unavailable time point was not counted as missing. The location-shift analysis held study weights and interval width fixed while applying one hypothetical shift to all missing observations. It identified the shift required for the pooled interval to touch zero.

Hallmark pathway analysis used singscore, with ssGSEA as a sensitivity method [6–8]. Pathway correlations used within-patient score changes and Benjamini-Hochberg correction [9]. Human Protein Atlas v25.1 annotations supplied broad blood-cell expression categories [10]. These labels were not treated as deconvolution.

Supplementary tables

Table S1. Cohort registry and analysis roles

Cohort	Population	Material	Platform	Baseline	Follow-up	Baseline/T24/T48 n	Role
GSE236713	Adult sepsis	PBL	Agilent GPL17077	Study day 1	T24	124/93/0	Primary longitudinal cohort
GSE57065	Adult septic shock	Whole blood	Affymetrix GPL570	H0	T24; T48	28/28/26	Primary longitudinal cohort
GSE95233	Adult septic shock	Whole blood	Affymetrix GPL570	Day 1	T24; T48	51/20/31	Primary longitudinal cohort
GSE54514	Adult sepsis	Whole blood	Illumina GPL6947	First ICU sample/day 1	T24; T48; extended	35/31/28	Primary longitudinal cohort
GSE110487	Adult septic shock	Whole blood	RNA-seq, GENCODE v25	Immediately after inclusion	T48	31/0/31	Primary longitudinal cohort
GSE8121	Pediatric septic shock	Whole blood	Affymetrix GPL570	PICU day 1	T48	30/0/30	Primary longitudinal cohort
GSE106878	Adult septic shock	Whole blood	Illumina GPL10295	Pre-treatment	T24	47/47/NA	Separate T24 directional-replication cohort

Baseline counts denote patients with a recoverable baseline sample; the source series contained 126 unique case IDs for GSE236713 and 36 for GSE54514.

Table S2. Fixed signature registry and implementation

ID	Signature	Original intended use	Genes	Executed formula	Oriented direction	Source DOI
SIG001	Sepsis MetaScore	Sepsis/infection vs noninfectious inflammation	11	$GM(CEACAM1, ZDHHC19, NMRK1, GNA15, BATF, C3AR1) - (5/6) \times GM(FAM214A, TGFB1, MTCH1, RPGRIP1, HLA-DPB1)$	Published positive direction retained	10.1126/scitranslmed.aaa5993
SIG002	SeptiCytte LAB	Sepsis vs infection-negative systemic inflammation	4	$PLAC8 + LAMP1 - PLA2G7 - CEACAM4$ on log/VST-like normalized expression	Published positive direction retained	10.1371/journal.pmed.1001916
SIG003	<i>FAIM3:PLAC8</i> ratio	CAP/infection vs noninfectious ICU mimics	2	$FAIM3 / PLAC8$; multiplied by -1 once for sepsis-positive orientation	Multiplied by -1 once so higher indicates infection/sepsis	10.1164/rccm.201502-0355OC
SIG004	sNIP	Abdominal sepsis vs postoperative inflammation	3	$(NLRP1 - IDNK) / PLAC8$	Published positive direction retained	10.1164/rccm.201707-1339LE
SIG022	Bacterial/Viral MetaScore	Bacterial vs viral infection	7	$GM(viral\ 3) - GM(bacterial\ 4)$	Published positive direction retained	10.1126/scitranslmed.aaf7165
SIG023	Herberg Disease Risk Score	Bacterial vs viral infection	2	$FAM89A - IFI44L$	Published positive direction retained	10.1001/jama.2016.11236
SIG033	Lin seven-gene mortality score	Mortality prognosis	7	Published seven coefficients applied without refitting or intercept	Published positive direction retained	10.3389/fimmu.2023.1000431
SIG034	Severe-or-Mild score	Host-response severity	42	$[GM(M1) + GM(M2)] / [GM(M3) + GM(M4)]$ using the published 42 genes	Published positive direction retained	10.1016/j.immuni.2021.03.002

All formulas were applied without refitting. GM, geometric mean. The Bacterial/Viral MetaScore omitted only its common positive gene-count scalar; standardized changes, signs, ranks and exact standardized contributions are unchanged.

Table S3. Primary temporal meta-analysis and modified Hartung-Knapp sensitivity

ID	Signature	Landmark	k	n	Pooled ΔZ	Standard HK 95% CI	Modified HK 95% CI	I ² (%)	95% PI	Holm P	Modified-HK Holm P
SIG001	Sepsis MetaScore	T24	4	172	-0.379	-0.432, -0.325	-0.560, -0.197	0.0	-0.451, -0.306	0.002	0.042
SIG001	Sepsis MetaScore	T48	5	146	-0.855	-1.351, -0.359	-1.368, -0.342	87.4	-2.176, 0.466	0.044	0.049
SIG002	SeptiCytte LAB	T24	3	141	-0.071	-0.181, 0.039	-0.299, 0.157	0.0	-0.397, 0.255	0.220	0.626
SIG002	SeptiCytte LAB	T48	4	118	-0.689	-0.906, -0.472	-0.980, -0.397	0.0	-0.982, -0.395	0.014	0.034
SIG003	<i>FAIM3:PLAC8</i>	T24	3	141	-0.173	-0.269, -0.077	-0.505, 0.159	0.0	-0.455, 0.109	0.048	0.462
SIG003	<i>FAIM3:PLAC8</i>	T48	4	118	-0.899	-1.224, -0.574	-1.242, -0.556	0.0	-1.354, -0.444	0.019	0.029
SIG004	sNIP	T24	3	141	0.027	-0.378, 0.431	-0.390, 0.444	23.7	-1.656, 1.710	0.801	0.807
SIG004	sNIP	T48	4	118	0.697	0.332, 1.062	0.281, 1.114	31.2	-0.150, 1.544	0.044	0.052
SIG022	Bacterial/Viral MetaScore	T24	4	172	-0.057	-0.315, 0.202		49.5	-0.613, 0.499		
SIG022	Bacterial/Viral MetaScore	T48	5	146	0.139	-0.191, 0.469		82.2	-0.720, 0.998		
SIG023	Herberg two-gene DRS	T24	4	172	0.047	-0.523, 0.617		85.6	-1.533, 1.626		
SIG023	Herberg two-gene DRS	T48	5	146	-0.053	-0.274, 0.169		44.1	-0.520, 0.414		
SIG033	Lin mortality score	T24	4	172	-0.162	-0.358, 0.033		2.7	-0.427, 0.102		
SIG033	Lin mortality score	T48	5	146	-0.120	-0.365, 0.125		18.5	-0.468, 0.228		
SIG034	Severe-or-Mild score	T24	4	172	0.030	-0.128, 0.188		0.0	-0.183, 0.243		
SIG034	Severe-or-Mild score	T48	5	146	-0.070	-0.283, 0.142		18.9	-0.451, 0.310		

Blank modified-HK fields indicate that the sensitivity family was prespecified only for the four diagnostic signatures. PI, prediction interval.

Table S4. Patient-rank persistence

ID	Signature	Landmark	k	n	Pooled ρ	Standard HK 95% CI	Modified HK 95% CI	I ² (%)	Modified 95% PI	Median displacement (pp)	Holm P
SIG001	Sepsis MetaScore	T24	4	172	0.618	0.063, 0.881	0.054, 0.883	76.1	-0.792, 0.987	14.0	0.124
SIG001	Sepsis MetaScore	T48	5	146	0.589	0.145, 0.835	0.145, 0.835	78.0	-0.584, 0.965	18.0	0.069
SIG002	SeptiCytte LAB	T24	3	141	0.634	-0.189, 0.934	-0.234, 0.940	79.0	-1.000, 1.000	15.8	0.135
SIG002	SeptiCytte LAB	T48	4	118	0.476	0.263, 0.644	0.203, 0.680	0.0	0.095, 0.735	23.3	0.053

Supplementary tables

ID	Signature	Landmark	k	n	Pooled p	Standard HK 95% CI	Modified HK 95% CI	I ² (%)	Modified 95% PI	Median displacement (pp)	Holm P
SIG003	<i>FAIM3:PLAC8</i>	T24	3	141	0.552	0.452, 0.637	0.186, 0.783	0.0	-0.576, 0.956	14.1	0.101
SIG003	<i>FAIM3:PLAC8</i>	T48	4	118	0.431	0.211, 0.610	0.121, 0.665	0.0	0.002, 0.727	19.6	0.069
SIG004	sNIP	T24	3	141	0.611	-0.121, 0.913	-0.125, 0.913	68.9	-0.998, 1.000	14.8	0.135
SIG004	sNIP	T48	4	118	0.426	0.166, 0.631	0.110, 0.664	0.0	-0.010, 0.726	19.0	0.069
SIG022	Bacterial/Viral MetaScore	T24	4	172	0.727	0.378, 0.895	0.376, 0.895	61.9	-0.385, 0.978	10.8	
SIG022	Bacterial/Viral MetaScore	T48	5	146	0.676	0.421, 0.832	0.420, 0.832	42.5	0.065, 0.918	10.0	
SIG023	Herberg two-gene DRS	T24	4	172	0.699	0.387, 0.868	0.355, 0.877	50.8	-0.282, 0.966	13.6	
SIG023	Herberg two-gene DRS	T48	5	146	0.600	0.271, 0.804	0.265, 0.806	50.6	-0.210, 0.922	13.3	
SIG033	Lin seven-gene mortality score	T24	4	172	0.448	-0.506, 0.909	-0.508, 0.910	92.2	-0.988, 0.998	15.7	
SIG033	Lin seven-gene mortality score	T48	5	146	0.440	0.275, 0.580	0.211, 0.624	0.0	0.174, 0.646	20.7	
SIG034	Severe-or-Mild score	T24	4	172	0.527	-0.016, 0.830	-0.036, 0.836	76.3	-0.789, 0.978	14.8	
SIG034	Severe-or-Mild score	T48	5	146	0.599	0.082, 0.862	0.075, 0.864	82.3	-0.714, 0.979	18.0	

Correlations were synthesized after Fisher-z transformation. pp, percentile points. Holm correction applies only to the four diagnostic signatures within each landmark.

Table S5. Clinical-course and severity contrasts

Cohort	Contrast	Endpoint	Signature	Favorable/adverse n	β	HC3 95% CI	P	Holm P
GSE95233	Admission to D2	28-day survival	Sepsis MetaScore	Survivor 12 / Non-survivor 8	0.652	-0.068, 1.372	0.076	0.304
GSE95233	Admission to D2	28-day survival	SeptiCyte LAB	Survivor 12 / Non-survivor 8	0.270	-0.699, 1.239	0.585	0.990
GSE95233	Admission to D2	28-day survival	<i>FAIM3:PLAC8</i>	Survivor 12 / Non-survivor 8	0.354	-0.664, 1.372	0.495	0.990
GSE95233	Admission to D2	28-day survival	sNIP	Survivor 12 / Non-survivor 8	-0.526	-1.513, 0.461	0.296	0.888
GSE95233	Admission to D2/D3	28-day survival	Sepsis MetaScore	Survivor 34 / Non-survivor 17	1.011	0.556, 1.465	1.33e-05	5.30e-05
GSE95233	Admission to D2/D3	28-day survival	SeptiCyte LAB	Survivor 34 / Non-survivor 17	0.429	-0.151, 1.009	0.147	0.147
GSE95233	Admission to D2/D3	28-day survival	<i>FAIM3:PLAC8</i>	Survivor 34 / Non-survivor 17	1.089	0.423, 1.755	0.001	0.003

Supplementary tables

Cohort	Contrast	Endpoint	Signature	Favorable/adverse n	β	HC3 95% CI	P	Holm P
GSE95233	Admission to D2/D3	28-day survival	sNIP	Survivor 34 / Non-survivor 17	-1.161	-1.822, -0.500	5.74e-04	0.002
GSE95233	Admission to D3	28-day survival	Sepsis MetaScore	Survivor 22 / Non-survivor 9	1.280	0.698, 1.862	1.63e-05	6.52e-05
GSE95233	Admission to D3	28-day survival	SeptiCytE LAB	Survivor 22 / Non-survivor 9	0.549	-0.193, 1.291	0.147	0.147
GSE95233	Admission to D3	28-day survival	<i>FAIM3:PLAC8</i>	Survivor 22 / Non-survivor 9	1.642	0.812, 2.472	1.06e-04	3.17e-04
GSE95233	Admission to D3	28-day survival	sNIP	Survivor 22 / Non-survivor 9	-1.638	-2.501, -0.776	1.97e-04	3.94e-04
GSE110487	T48	source-defined organ-function course group	Sepsis MetaScore	R 17 / NR 14	-0.452	-0.963, 0.060	0.084	0.335
GSE110487	T48	source-defined organ-function course group	SeptiCytE LAB	R 17 / NR 14	-0.291	-1.026, 0.444	0.438	1.000
GSE110487	T48	source-defined organ-function course group	<i>FAIM3:PLAC8</i>	R 17 / NR 14	-0.127	-0.748, 0.495	0.690	1.000
GSE110487	T48	source-defined organ-function course group	sNIP	R 17 / NR 14	-0.043	-0.622, 0.537	0.885	1.000
GSE57065	T24	baseline SAPS II severity category	Sepsis MetaScore	SAPSII-Low 14 / SAPSII-High 14	0.750	0.220, 1.280	0.006	0.017
GSE57065	T24	baseline SAPS II severity category	SeptiCytE LAB	SAPSII-Low 14 / SAPSII-High 14	0.794	0.098, 1.490	0.025	0.034
GSE57065	T24	baseline SAPS II severity category	<i>FAIM3:PLAC8</i>	SAPSII-Low 14 / SAPSII-High 14	1.181	0.490, 1.872	8.06e-04	0.003
GSE57065	T24	baseline SAPS II severity category	sNIP	SAPSII-Low 14 / SAPSII-High 14	-0.799	-1.456, -0.142	0.017	0.034
GSE57065	T48	baseline SAPS II severity category	Sepsis MetaScore	SAPSII-Low 14 / SAPSII-High 12	0.857	0.195, 1.518	0.011	0.033
GSE57065	T48	baseline SAPS II severity category	SeptiCytE LAB	SAPSII-Low 14 / SAPSII-High 12	0.983	0.187, 1.779	0.016	0.033
GSE57065	T48	baseline SAPS II severity category	<i>FAIM3:PLAC8</i>	SAPSII-Low 14 / SAPSII-High 12	1.382	0.607, 2.157	4.74e-04	0.002
GSE57065	T48	baseline SAPS II severity category	sNIP	SAPSII-Low 14 / SAPSII-High 12	-0.774	-1.614, 0.066	0.071	0.071
GSE54514	T24	source-defined survival status	Sepsis MetaScore	Survivor 24 / Non-survivor 7	-0.572	-1.538, 0.393	0.245	0.764
GSE54514	T24	source-defined survival status	SeptiCytE LAB	Survivor 24 / Non-survivor 7	0.321	-0.572, 1.215	0.481	0.764
GSE54514	T24	source-defined survival status	<i>FAIM3:PLAC8</i>	Survivor 24 / Non-survivor 7	-0.720	-1.798, 0.359	0.191	0.764
GSE54514	T24	source-defined survival status	sNIP	Survivor 24 / Non-survivor 7	0.521	-0.563, 1.604	0.346	0.764

Supplementary tables

Cohort	Contrast	Endpoint	Signature	Favorable/adverse n	β	HC3 95% CI	P	Holm P
GSE54514	T48	source-defined survival status	Sepsis MetaScore	Survivor 21 / Non-survivor 7	0.463	-0.588, 1.514	0.388	1.000
GSE54514	T48	source-defined survival status	SeptiCytE LAB	Survivor 21 / Non-survivor 7	0.660	-0.280, 1.599	0.169	0.674
GSE54514	T48	source-defined survival status	<i>FAIM3:PLAC8</i>	Survivor 21 / Non-survivor 7	-0.353	-1.479, 0.772	0.538	1.000
GSE54514	T48	source-defined survival status	sNIP	Survivor 21 / Non-survivor 7	-0.209	-1.320, 0.902	0.712	1.000

β is the adverse-minus-favorable difference in paired ΔZ . D2 and D3 are sensitivity contrasts; the combined GSE95233 model adjusts for exact follow-up day. Holm correction was applied across the four diagnostic signatures within each dataset and contrast.

Table S6. Diagnostic-signature change concordance

Landmark	Pair	Signatures	k	n	Pooled p	Modified HK 95% CI	Modified 95% PI	I ² (%)	Median sign agreement	Holm P
T24	SIG001 / SIG002	Sepsis MetaScore / SeptiCytE LAB	3	141	0.378	-0.266, 0.788	-0.990, 0.998	37.8	0.65	0.627
T24	SIG001 / SIG003	Sepsis MetaScore / <i>FAIM3:PLAC8</i>	3	141	0.632	-0.685, 0.981	-1.000, 1.000	83.6	0.70	0.627
T24	SIG001 / SIG004	Sepsis MetaScore / sNIP	3	141	-0.441	-0.856, 0.319	-1.000, 0.999	69.9	0.42	0.627
T24	SIG002 / SIG003	SeptiCytE LAB / <i>FAIM3:PLAC8</i>	3	141	0.280	-0.400, 0.762	-0.996, 0.999	55.7	0.55	0.627
T24	SIG002 / SIG004	SeptiCytE LAB / sNIP	3	141	-0.412	-0.864, 0.407	-1.000, 0.999	72.8	0.35	0.627
T24	SIG003 / SIG004	<i>FAIM3:PLAC8</i> / sNIP	3	141	-0.393	-0.750, 0.142	-0.994, 0.969	35.6	0.32	0.512
T48	SIG001 / SIG002	Sepsis MetaScore / SeptiCytE LAB	4	118	0.599	0.377, 0.755	0.285, 0.797	0.0	0.80	0.030
T48	SIG001 / SIG003	Sepsis MetaScore / <i>FAIM3:PLAC8</i>	4	118	0.619	0.309, 0.810	-0.157, 0.922	40.0	0.80	0.045
T48	SIG001 / SIG004	Sepsis MetaScore / sNIP	4	118	-0.584	-0.820, -0.178	-0.955, 0.502	61.4	0.21	0.045
T48	SIG002 / SIG003	SeptiCytE LAB / <i>FAIM3:PLAC8</i>	4	118	0.536	0.277, 0.723	0.171, 0.772	0.0	0.73	0.045
T48	SIG002 / SIG004	SeptiCytE LAB / sNIP	4	118	-0.526	-0.768, -0.152	-0.902, 0.304	33.5	0.20	0.045
T48	SIG003 / SIG004	<i>FAIM3:PLAC8</i> / sNIP	4	118	-0.673	-0.852, -0.352	-0.943, 0.136	39.4	0.17	0.045

Supplementary tables

Positive correlations indicate parallel within-patient change. A negative correlation indicates opposing movement. Holm correction covered all six signature pairs within each landmark.

Table S7. GSE106878 T24 directional replication and matched-gene analysis

ID	Signature	T24 ΔZ (bootstrap 95% CI)	Rank ρ (bootstrap 95% CI)	Absolute-effect null percentile	Empirical P	Treatment-adjusted mortality β (95% CI)	Mortality Holm P
SIG001	Sepsis MetaScore	-0.972 (-1.139, -0.815)	0.822 (0.708, 0.896)	0.985	0.016	0.300 (-0.124, 0.723)	0.661
SIG002	SeptiCyt LAB	-1.484 (-1.776, -1.201)	0.403 (0.114, 0.641)	1.000	9.99e-04	0.309 (-0.201, 0.819)	0.705
SIG003	<i>FAIM3:PLAC8</i>	-1.117 (-1.351, -0.877)	0.581 (0.343, 0.750)	0.996	0.005	0.301 (-0.218, 0.820)	0.705
SIG004	sNIP	0.917 (0.716, 1.124)	0.707 (0.523, 0.823)	0.992	0.009	-0.173 (-0.642, 0.297)	0.705
SIG022	Bacterial/Viral MetaScore	0.183 (0.028, 0.339)	0.715 (0.519, 0.840)	0.393	0.607	-0.040 (-0.353, 0.273)	
SIG023	Herberg two-gene DRS	-0.208 (-0.368, -0.025)	0.704 (0.455, 0.874)	0.493	0.507	0.271 (-0.138, 0.680)	
SIG033	Lin mortality score	0.208 (-0.036, 0.452)	0.584 (0.326, 0.779)	0.431	0.569	-0.044 (-0.612, 0.523)	
SIG034	Severe-or-Mild score	0.187 (-0.046, 0.427)	0.648 (0.396, 0.805)	0.370	0.630	0.035 (-0.497, 0.567)	

The null used 1,000 formula-preserving replacements per signature, matched exactly on baseline mean and variance deciles. Mortality β compares nonsurvivors with survivors while adjusting for randomized treatment arm.

Table S8. Exact contribution architecture and recurrence at T48

ID	Signature	Drift architecture	Median dominance	Median cancellation	Pooled leading gene	Leading absolute share	Recurring cohort leader	Recurrence
SIG001	Sepsis MetaScore	Cohort Dependent Drift	0.291	0.295	<i>CEACAM1</i>	0.307	<i>CEACAM1</i>	3/5
SIG002	SeptiCyt LAB	Consistent Multigene Drift	0.510	0.490	<i>PLAC8</i>	0.597	<i>PLAC8</i>	4/4
SIG003	<i>FAIM3:PLAC8</i>	Single Gene Dominant Drift	0.679	0.000	<i>PLAC8</i>	0.599	<i>PLAC8</i>	4/4
SIG004	sNIP	Consistent Multigene Drift	0.589	0.345	<i>NLRP1</i>	0.485	<i>NLRP1</i>	3/4
SIG022	Bacterial/Viral MetaScore	Cohort Dependent Drift	0.389	0.507	<i>HK3</i>	0.278	<i>IFI27</i>	3/5
SIG023	Herberg two-gene DRS	Single Gene Dominant Drift	0.735	0.000	<i>FAM89A</i>	0.754	<i>FAM89A</i>	4/5
SIG033	Lin mortality score	Cohort Dependent Drift	0.339	0.610	<i>LTB</i>	0.288	<i>LTB</i>	3/5
SIG034	Severe-or-Mild score	Cohort Dependent Drift	0.098	0.773	<i>DEFA4</i>	0.097	<i>DEFA4</i>	2/5

All 3,344 patient-signature-window contribution sums reproduced total ΔZ within 1×10^{-8} ; the maximum observed error was 1.72×10^{-14} . Contribution labels describe formula arithmetic rather than causality.

Table S9A. Landmark follow-up availability

Cohort	Landmark	Baseline eligible	Observed pairs	Missing	Missing (%)	Reason availability
GSE236713	T24	124	93	31	25.0	All patient-level missing reasons unavailable in public metadata
GSE57065	T24	28	28	0	0.0	All patient-level missing reasons unavailable in public metadata
GSE57065	T48	28	26	2	7.1	All patient-level missing reasons unavailable in public metadata
GSE95233	T24	51	20	31	60.8	All patient-level missing reasons unavailable in public metadata
GSE95233	T48	51	31	20	39.2	All patient-level missing reasons unavailable in public metadata
GSE54514	T24	35	31	4	11.4	All patient-level missing reasons unavailable in public metadata
GSE54514	T48	35	28	7	20.0	All patient-level missing reasons unavailable in public metadata
GSE110487	T48	31	31	0	0.0	All patient-level missing reasons unavailable in public metadata
GSE8121	T48	30	30	0	0.0	All patient-level missing reasons unavailable in public metadata

Landmarks that were not scheduled by a source study were not classified as missing follow-up.

Table S9B. Location-shift missingness sensitivity for diagnostic signatures

ID	Signature	Landmark	Observed ΔZ (95% CI)	CI excludes zero	Absolute missing-observed shift for CI to touch zero
SIG001	Sepsis MetaScore	T24	-0.379 (-0.432, -0.325)	Yes	1.327
SIG001	Sepsis MetaScore	T48	-0.855 (-1.351, -0.359)	Yes	2.861
SIG002	SeptiCytte LAB	T24	-0.071 (-0.181, 0.039)	No	
SIG002	SeptiCytte LAB	T48	-0.689 (-0.906, -0.472)	Yes	5.249
SIG003	<i>FAIM3:PLAC8</i>	T24	-0.173 (-0.269, -0.077)	Yes	0.292
SIG003	<i>FAIM3:PLAC8</i>	T48	-0.899 (-1.224, -0.574)	Yes	6.472
SIG004	sNIP	T24	0.027 (-0.378, 0.431)	No	
SIG004	sNIP	T48	0.697 (0.332, 1.062)	Yes	3.430

The sensitivity holds study weights and interval width fixed. It identifies a hypothetical common missing-versus-observed ΔZ shift and is not an imputation model.

Table S10. Clinical metadata provenance

Dataset	Variable	Serial timing	Participants/observations	Mapping reliability	Role
GSE54514	Survival status	Day 1 (baseline), Day 2 (~24 h), and Day 3 (~48 h); later Day 4 and Day 5 samples were available but were not required for the early paired contrasts	35 patients at baseline; 31 baseline-to-Day-2 pairs (62 observations) and 28 baseline-to-Day-3 pairs (56 observations)	High for the baseline-defined paired sets after exclusion of the isolated Day-4 record	Secondary source-defined survival context

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Dataset	Variable	Serial timing	Participants/observations	Mapping reliability	Role
GSE95233	Day-28 survival	Admission/Day 1 and a source-defined second sample on Day 2 (n=20) or Day 3 (n=31)	51 patients with complete pairs (102 observations): 34 survivors and 17 nonsurvivors	High; all 51 patient pairs and outcome labels were resolved	Primary mortality clinical-course anchor
GSE110487	Early organ-function response	T1 within 16 h of ICU admission and T2 48 h after study enrolment	31 patients with complete pairs (62 observations): 17 responders and 14 nonresponders	High after reconciliation; a separate GEO timepoint characteristic disagreed with the concordant identifiers in 21 of 62 records and was not used	Primary organ-function-response clinical-course anchor
GSE57065	Baseline SAPS II severity class	H0 within 30 min of vasopressor initiation, H24, and H48	28 H0-to-H24 pairs (56 observations) and 26 H0-to-H48 pairs (52 observations); 14 SAPSII-Low and 14 SAPSII-High patients at baseline	High for time and the binary SAPS II class; exact patient-level SAPS II values and mortality assignments were not required or inferred	Secondary severity-trajectory context
GSE106878	Treatment arm and day-28 survival	Pre-treatment and 24 h post-treatment	47 patients with complete pairs (94 observations): 24 hydrocortisone and 23 placebo; 34 day-28 survivors and 13 nonsurvivors	High; 47 complete pairs were resolved and all 74 genes required by the eight fixed formulas were covered	Separate T24 directional replication outside the primary synthesis; treatment-aware mortality context is secondary.

Table S11. Public longitudinal dataset search-update and eligibility decisions

Dataset	Source	Population/design	Repeated samples	Public expression input	Gene coverage	Final role	Decision/reason
GSE106878	GEO	Adult septic shock; pre-treatment and 24 h	47 complete pairs	Processed series matrix	74/74 required genes	Separate directional replication	Included outside primary synthesis
E-MEXP-3850	BioStudies/ArrayExpress	Five children; six samples over 0–48 h	Yes	No raw or processed expression matrix	Not assessable	Not analyzed	Excluded: expression input unavailable
PRJEB111201	ENA	Eleven adults; whole-blood RNA-seq on days 1, 3 and 7	10 complete D1/D3/D7 series; one D1 only	Raw FASTQ only; no processed subject-by-gene matrix	Not verifiable on established scale	Not analyzed	Excluded: new raw-read workflow would be required after analysis freeze
phs003608.v1.p1	dbGaP	Ghana whole-blood RNA-seq at 0, 6, 24, 48 and 72 h	Yes	Subject-level bulk matrix controlled access	Not verifiable from open files	Not analyzed	Excluded: no openly accessible complete subject-by-gene bulk matrix

The four rows are the candidates formally adjudicated in the 20 July 2026 update. The broader search registry, source URLs and dated decision records are retained in S1 Data.

Table S12. Published signature candidate registry and reproducibility eligibility decisions

Signature	Intended use	Gene type	Grade	Fixed formula available	Direction defined	Cross-platform mapping	No-refit scoring	Final inclusion
SIG001 Sepsis MetaScore	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG002 SeptiCytte LAB	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG003 FAIM3:PLAC8 ratio	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG004 sNIP score	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG005 Bauer seven-gene expression score	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Yes; equation incomplete	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG006 SeptiCytte RAPID	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	X	No public locked implementation	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade X; see source reason
SIG007 LIFTS	Diagnostic discovery with healthy controls	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG008 Tang PBMC 138-gene SVM signature	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG009 Sutherland 42-marker LogitBoost test	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG010 SepSigLnc	Diagnostic discovery with healthy controls	lncRNA	A2	Yes; fixed ordering rule	Yes	Unresolved across required platforms	No	Not evaluated: lncRNA mapping unresolved
SIG011 Nine-mRNA random-forest signature	Diagnostic discovery with healthy controls	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG012 ARG classifier	Diagnostic/prognostic	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason

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Signature	Intended use	Gene type	Grade	Fixed formula available	Direction defined	Cross-platform mapping	No-refit scoring	Final inclusion
SIG013 ANEMONES SIRS-or-Sepsis five-gene signature	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG014 ANEMONES inflammation two-gene ratio	Host response	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG015 scPAGE2 sepsis model	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG016 EARLI whole-blood host bSVM classifier	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG017 Presymptomatic postoperative infection/sepsis signatures	Infection/sepsis detection	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG018 Sepset	Prognosis	Protein-coding RNA panel/model	X	No public locked implementation	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade X; see source reason
SIG019 28-gene sepsis-versus-healthy signature	Diagnostic discovery with healthy controls	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG020 13-gene sepsis-versus-SIRS signature	Sepsis/infection vs noninfectious inflammation	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG021 InSep IMX-BVN-2	Bacterial vs viral infection	Protein-coding RNA panel/model	X	No public locked implementation	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade X; see source reason
SIG022 Bacterial/Viral MetaScore	Bacterial vs viral infection	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG023 Herberg two-transcript bacterial/viral signature	Bacterial vs viral infection	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG024 Tsalik acute respiratory illness host classifiers	Infection vs noninfection	Protein-coding RNA panel/model	X	No public locked implementation	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade X; see source reason
SIG025 IMX-BVN-1	Bacterial vs viral infection	Protein-coding RNA panel/model	X	No public locked implementation	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade X; see source reason
SIG026 Ramilo 35-gene influenza-versus-bacterial classifier	Bacterial vs viral infection	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG027 Sepsis Response Signature classifier	Molecular subtype	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG028 MARS 140-gene endotype signature	Molecular subtype	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG029 SRSq / Sepstratifier	Prognosis/subtype	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG030 Stanford 12-gene sepsis mortality score	Prognosis	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Not separately established	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason

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Signature	Intended use	Gene type	Grade	Fixed formula available	Direction defined	Cross-platform mapping	No-refit scoring	Final inclusion
SIG031 Duke 18-gene sepsis mortality score	Prognosis	Protein-coding RNA panel/model	B	Complete executable rule not established in audit	Conflicting source directions	Not assessed after rule exclusion	Not established	Not included: grade B; see source reason
SIG032 SepSig28	Diagnostic discovery with healthy controls	lncRNA	A2	Yes; fixed ordering rule	Yes	Unresolved across required platforms	No	Not evaluated: lncRNA mapping unresolved
SIG033 Lin seven-immune-gene mortality score	Prognosis	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included
SIG034 Severe-or-Mild score	Host-response severity	Protein-coding RNA panel/model	A2	Yes; fixed ordering rule	Yes	Resolved	Yes	Included

Grades and intended-use strata were frozen on 15 July 2026 before longitudinal outcomes were inspected. A2 denotes a recoverable fixed ordering score, not a transportable assay scale or clinical cutoff. B/X candidates did not meet the audited fixed-rule requirements; where direction or mapping was not separately established, this is an audit limitation rather than proof of impossibility. The full source locator, evidence status and individual grade reason for every candidate are retained in S1 Data sheet S2_SelectionAudit.

Table S13A. GSE106878 treatment-arm sensitivity for the four diagnostic signatures

Signature	Placebo n; mean ΔZ (bootstrap 95% CI)	Hydrocortisone n; mean ΔZ (bootstrap 95% CI)	β (H-P)	HC3 SE	HC3 95% CI	Raw P	Holm P	R ²	Same sign
Sepsis MetaScore	23; -0.977 (-1.189, -0.762)	24; -0.968 (-1.223, -0.717)	0.009	0.175	-0.334, 0.352	0.960	0.960	0.000	Yes
SeptiCytE LAB	23; -0.833 (-1.087, -0.593)	24; -2.107 (-2.466, -1.739)	-1.274	0.236	-1.736, -0.812	6.55e-8	2.62e-7	0.400	Yes
FAIM3:PLAC8	23; -0.838 (-1.126, -0.545)	24; -1.384 (-1.724, -1.042)	-0.546	0.244	-1.024, -0.068	0.025	0.076	0.103	Yes
sNIP	23; 0.811 (0.534, 1.070)	24; 1.019 (0.709, 1.317)	0.207	0.218	-0.219, 0.634	0.340	0.681	0.021	Yes

Arm-specific confidence intervals are percentile bootstrap intervals from 2,000 replicates. H-P denotes hydrocortisone minus placebo. β compares hydrocortisone with placebo under $\Delta Z = \beta_0 + \beta_1(\text{hydrocortisone})$; HC3 large-sample normal inference is shown. Holm correction covers the four β_1 tests. Directional sensitivity is not an efficacy analysis.

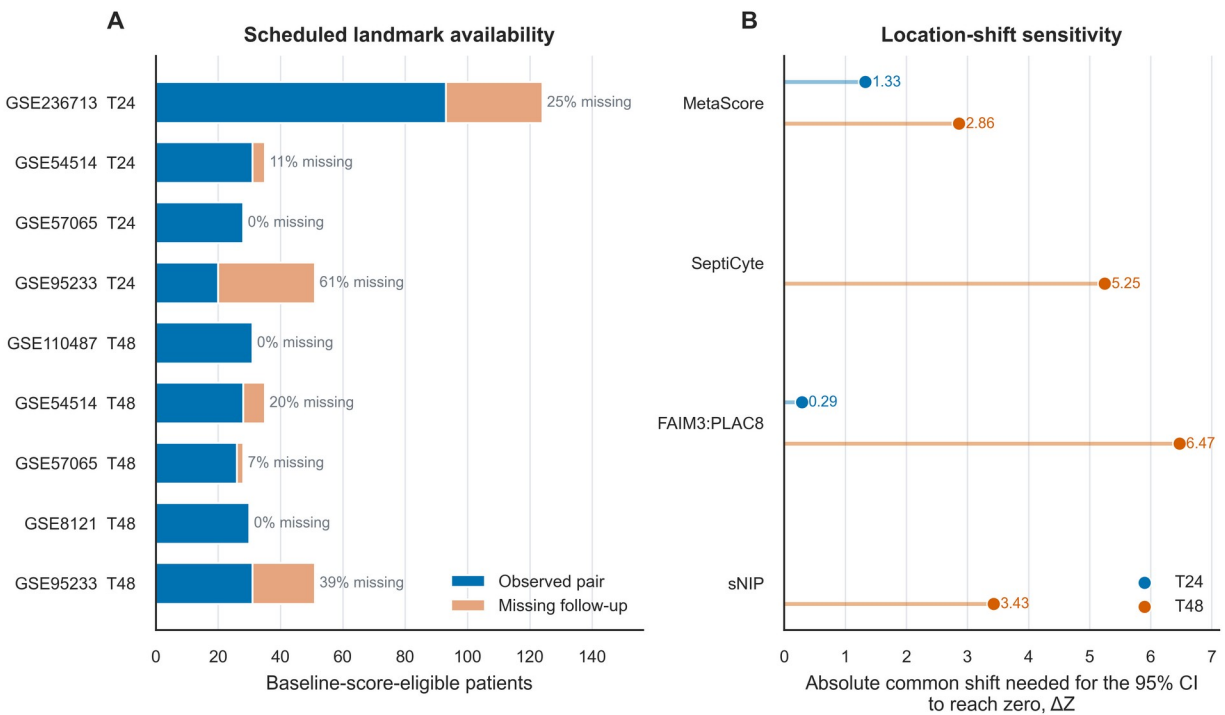
Table S13B. Complete coefficient output for the GSE106878 treatment-arm models

Signature	Term	Estimate	HC3 SE	95% CI	Raw P	Holm P	n	R ²
Sepsis MetaScore	Intercept	-0.977	0.112	-1.197, -0.756	3.78e-18	NA	47	0.000
Sepsis MetaScore	hydrocortisone	0.009	0.175	-0.334, 0.352	0.960	0.960	47	0.000
SeptiCytE LAB	Intercept	-0.833	0.138	-1.103, -0.563	1.41e-9	NA	47	0.400
SeptiCytE LAB	hydrocortisone	-1.274	0.236	-1.736, -0.812	6.55e-8	2.62e-7	47	0.400
FAIM3:PLAC8	Intercept	-0.838	0.155	-1.142, -0.534	6.79e-8	NA	47	0.103
FAIM3:PLAC8	hydrocortisone	-0.546	0.244	-1.024, -0.068	0.025	0.076	47	0.103
sNIP	Intercept	0.811	0.146	0.525, 1.098	2.90e-8	NA	47	0.021
sNIP	hydrocortisone	0.207	0.218	-0.219, 0.634	0.340	0.681	47	0.021

The intercept equals the placebo mean ΔZ . Holm-adjusted P values apply only to the four hydrocortisone coefficients. NA, not applicable: Holm correction was not applied to intercepts. Model R² is the ordinary least-squares goodness-of-fit measure; robust covariance affects inference, not the fitted values.

Supplementary figure legends

Figure S1. Landmark-specific follow-up availability and missingness sensitivity. Observed and missing paired samples are shown for scheduled T24 and T48 landmarks. The companion panel shows the hypothetical common shift among missing observations required for an initially nonzero confidence interval to reach zero.



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Figure S2. Cohort-level patient-rank persistence. Baseline-to-follow-up Spearman correlations and bootstrap 95% confidence intervals are shown for every eligible cohort-signature-landmark combination with at least 10 patients. Cohorts excluded because of development overlap remain identifiable in the source table and do not enter the pooled estimate.

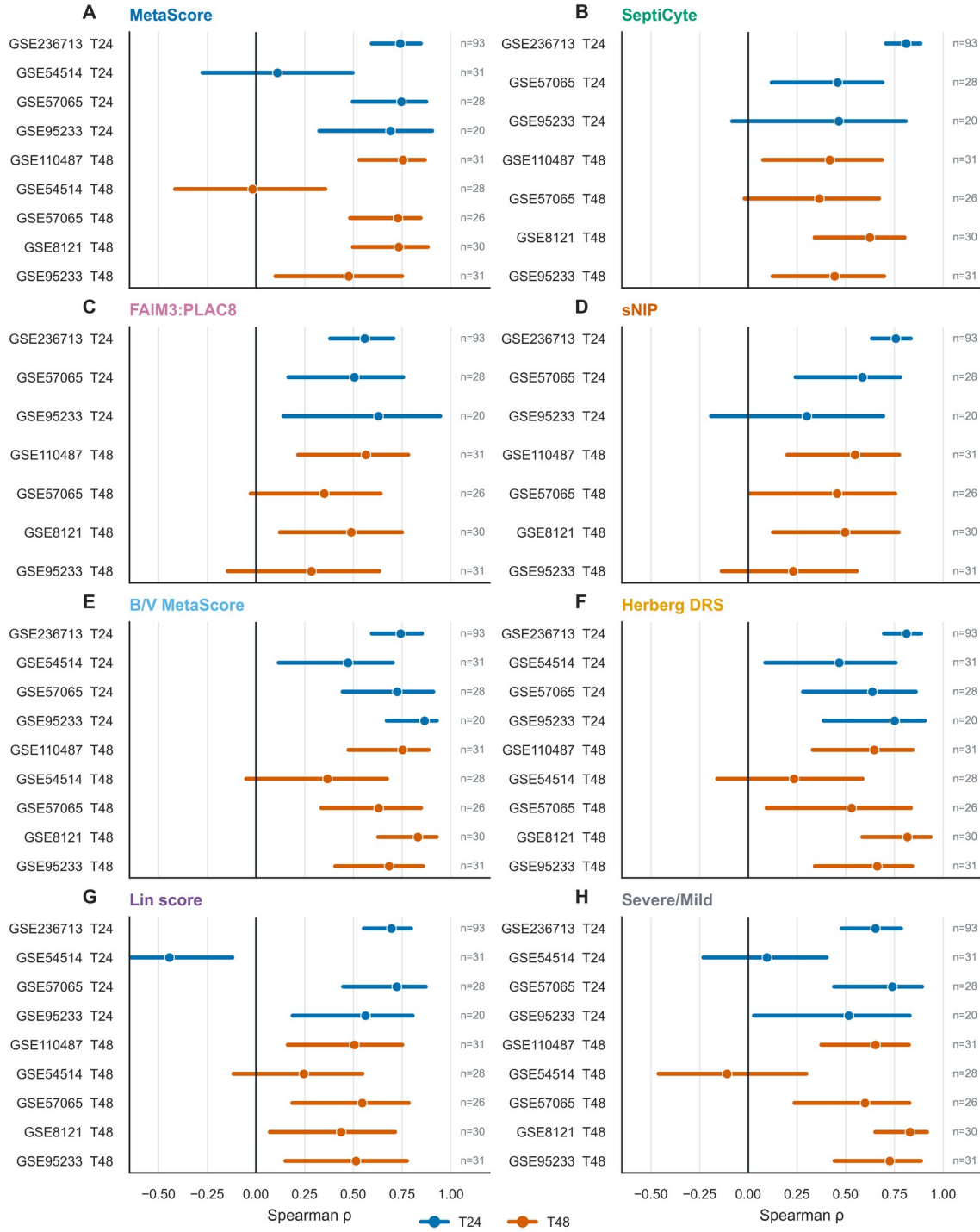


Figure S3. Standardized clinical-group trajectories. Mean baseline-standardized scores and bootstrap 95% confidence intervals are shown for GSE95233 survival, GSE110487 organ-function course, GSE57065 baseline SAPS II category and GSE54514 source-defined survival. Sample sizes are shown at each source-defined time.

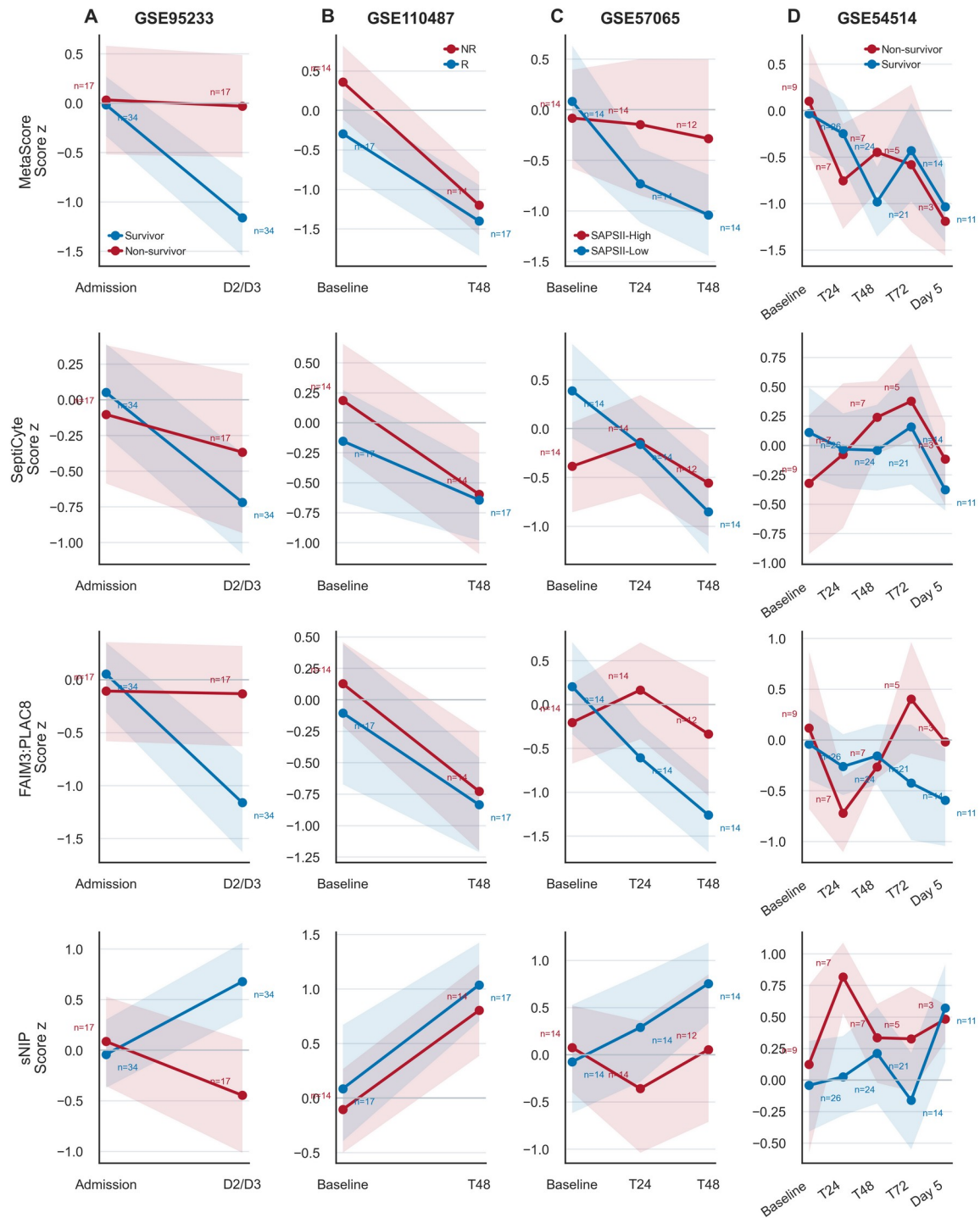


Figure S4. GSE106878 secondary clinical and rank analyses. Treatment-adjusted adverse-minus-favorable 28-day mortality contrasts are shown beside baseline-to-T24 rank-persistence estimates. These analyses use the same 47 patients as the separate T24 directional-replication cohort and do not represent an additional cohort.

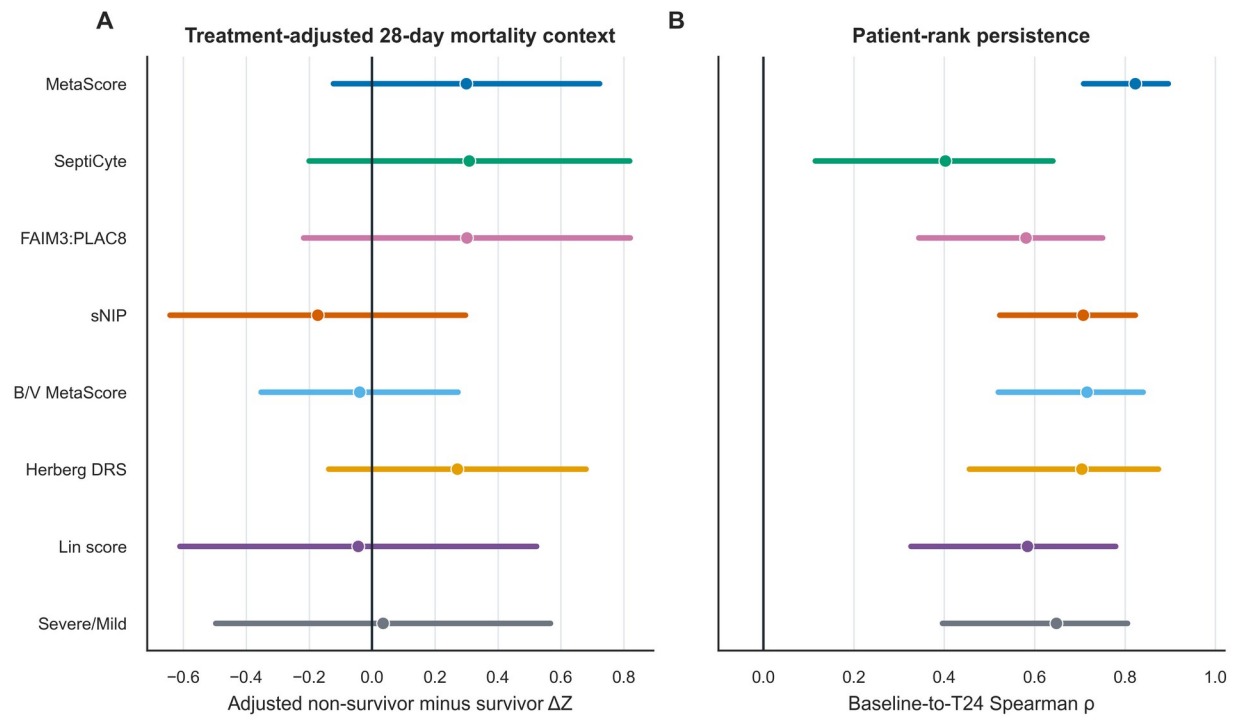


Figure S5. Matched-gene benchmark distributions in GSE106878. Distributions contain 1,000 baseline mean- and variance-matched pseudo-formulas per signature. Colored points show observed GSE106878 effects. Empirical two-sided P values and exact-bin matching rates are provided in the panel annotation and S1 Data.

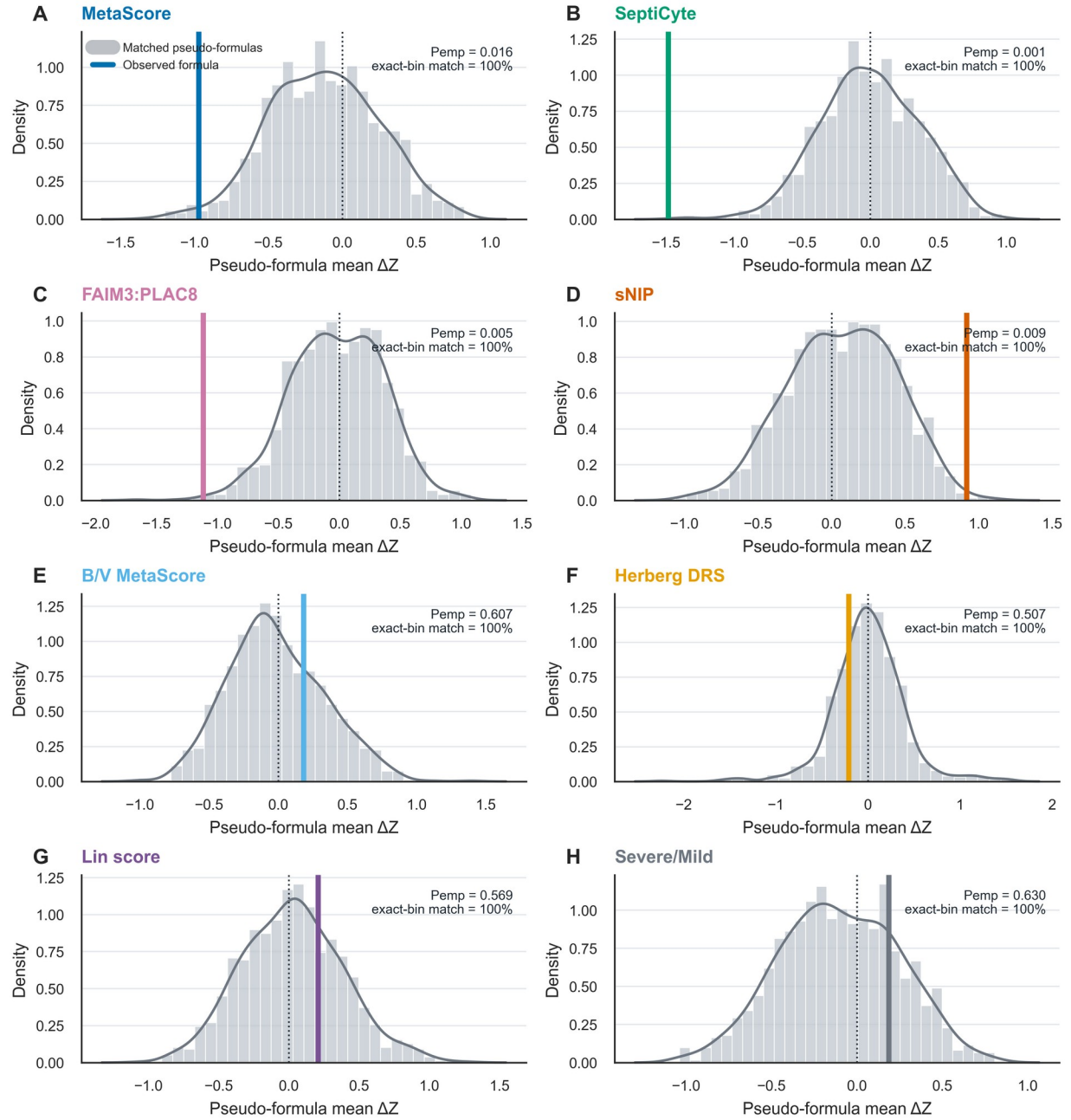
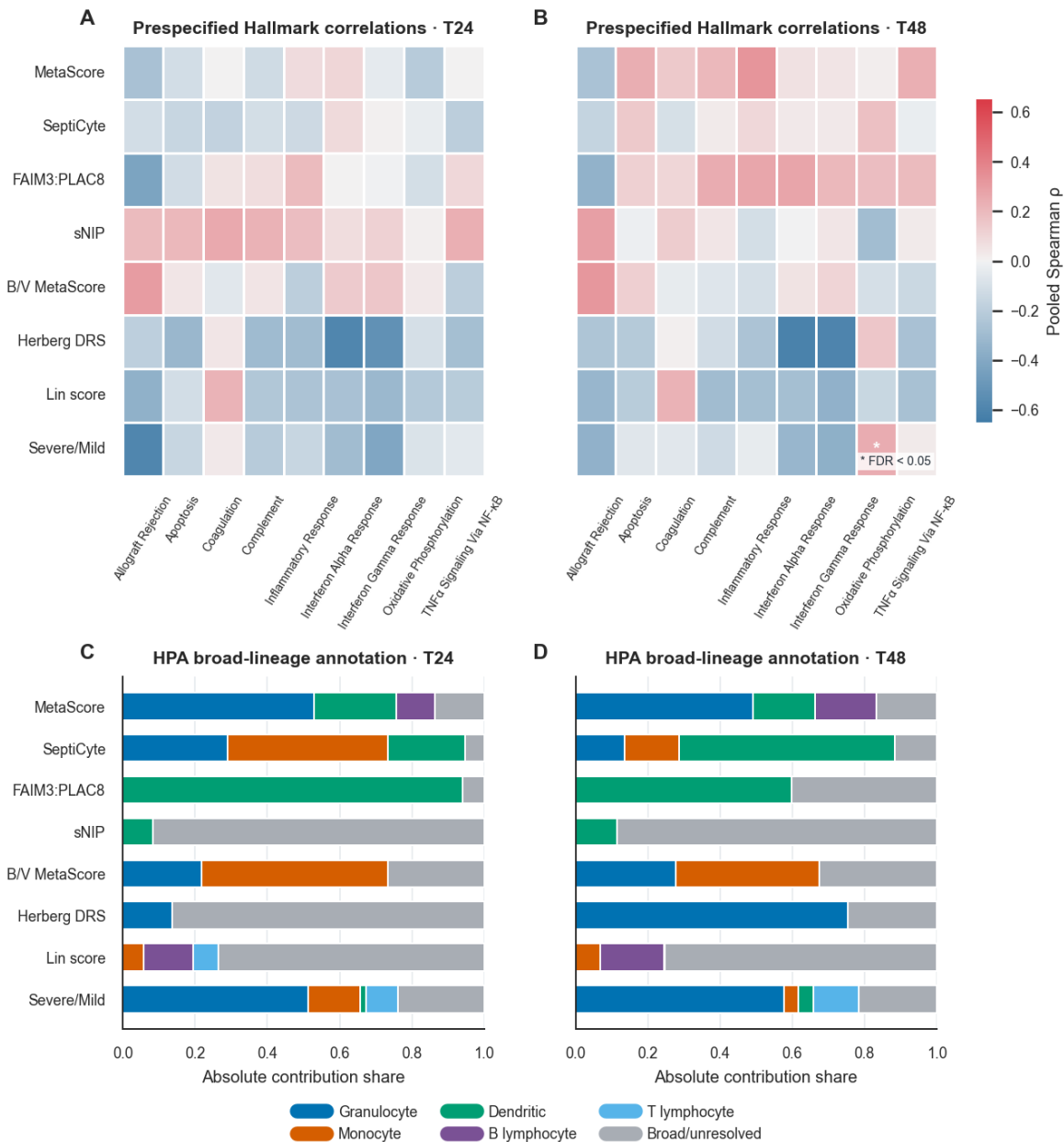


Figure S6. Supplementary biological context. Pooled signature-pathway change correlations are shown for prespecified Hallmark sets. The accompanying panel summarizes the absolute mathematical contribution assigned to broad Human Protein Atlas blood-cell expression categories. These annotations do not identify causal cell states.



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